

Appointments

- Assistant Professor. Department of Applied Mathematics and Statistics. Colorado School of Mines. August 2021 - present
- Assistant Adjunct Professor. Department of Mathematics, University of California, Los Angeles. July 2019 - August 2021
- Givens Associate. MCS Division, Argonne National Laboratory. May 15, 2018 - Nov 30, 2018

Education

- PhD. in Applied Mathematics, Emory University, Atlanta, GA, May 2019
Advisor: Lars Ruthotto
- BSc. in Applied Mathematics, Brown University, Providence, RI, May 2014
Advisor: Johnny Guzmán
- AA. in Mathematics, Miami Dade College, Miami, FL, May 2011

Research Interests

- Inverse Problems, Deep Learning, Optimization, Optimal Control, Mean Field Games

Funding

- NSF DMS 2309810: Optimization-based Implicit Deep Learning, Theory and Applications, funded by the US National Science Foundation. Total budget \$294,995. Principal Investigator. June 2023 - May 2026.
- NSF DMS 2110745*: Development of Geometrically-Flexible Physics-Based Convolution Kernels, funded by the US National Science Foundation. Total budget \$297,627. Principal Investigator. June 2021 - May 2025.
- EEMDL: Energy Emission Modeling Data Lab, funded by Cheniere Energy Inc. Total budget \$374,835. co-Principal Investigator. December 2025 - November 2026.

Preprints/Submitted Articles

1. Cherry C, Wu Fung S, Tenorio L, Bozdog E. A Gauss-Newton Method with No Additional PDE Solves Beyond Gradient Evaluation for Large-Scale PDE-Constrained Inverse Problems. arXiv:2603.06899, 2026

*Originally awarded to (Jennifer K Ryan) and transferred to (Samy Wu Fung) upon Dr. Ryan's departure from the Mines.

2. Gelphman E, Verma D, Yang NT, Osher S, Wu Fung S. On the Convergence of Jacobian-Free Backpropagation for Optimal Control Problems with Implicit Hamiltonians. *arXiv:2602.00921*, 2026
3. Zhang Z, Wu Fung S, Kyrillidis A, Osher S, Vardi M. Thinking Out of the Box: Hybrid SAT Solving by Unconstrained Continuous Optimization. *arXiv:2506.00674*, 2025

Published/Accepted Articles (Peer Reviewed)

1. Tomaselli C, Maffettone GC, Wu Fung S, Nurbekyan L, di Bernardo M. Mean-field control barrier functions for stochastic multi-agent systems. *IEEE Control Systems Letters* 2026
2. Gal E, Wu Fung S, Haber E. Probabilistic Gaussian Homotopy: A Probability-Space Continuation Framework for Nonconvex Optimization. *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases. (ECML-PKDD)*, 2026
3. Di N, Chi E, Wu Fung S. Operator Splitting with Hamilton-Jacobi-based Proximals. *International Conference on Machine Learning (ICML)*, 2026
4. Wu Fung S, Berkels B. A Generalization Bound for a Family of Implicit Networks. *Neurocomputing*, 2026.
5. Gelphman E, Verma D, Yang NT, Osher S, Wu Fung S. End-to-End Training of High-Dimensional Optimal Control with Implicit Hamiltonians via Jacobian-Free Backpropagation. *American Control Conference (ACC26)*, 2026
6. Meng T, Liu S, Wu Fung S, Osher S. Recent Advances in Numerical Solutions for Hamilton-Jacobi PDEs. *Communications on Applied Mathematics and Computation*, 2025.
7. Knutson B, Chyba A, Ivanitskiy M, Pettyjohn J, Diniz-Behn C, McKenzie D, Wu Fung S. On Logical Extrapolation for Mazes with Recurrent and Implicit Networks. *AAAI Conference on Artificial Intelligence*, 2025.
8. Wang X, Wu Fung S, Nurbekyan L. A Primal-Dual Price-Optimization Method for Computing Equilibrium Prices in Mean-Field Games Models. *Dynamic Games and Applications*, 2025
9. Ivanitskiy, MI, Spies AF, Räuker T, Knutson B, Diniz-Behn C, Wu Fung S. maze-dataset: Maze Generation with Algorithmic Variety and Representational Flexibility. *Journal of Open Source Software*, 2025.
10. Terrab S, Wu Fung S, Ryan JK. A hybrid SIAC–data-driven post-processing filter for discontinuities in solutions to numerical PDEs. *Journal of Engineering Mathematics*, 2025
11. Parada R, Wu Fung S, Osher S. Fast Partial Fourier Transforms for Large-Scale Ptychography. *Inverse Problems and Imaging*, 2025
12. Tibshirani R, Wu Fung S, Heaton H, Osher S. Laplace Meets Moreau: Smooth and Efficient Approximations to Infimal Convolutions Using Laplace's Method. *Journal of Machine Learning Research (JMLR)*, 2025

13. Wu Fung S, Nurbekyan L. Mean-Field Control Barrier Functions: A Framework for Real-Time Swarm Control. *American Control Conference (ACC25)*, 2025
14. Vidal A, Wu Fung S, Osher S, Tenorio L, Nurbekyan L. Kernel Expansions for High-Dimensional Mean Field Control with Nonlocal Interactions, *American Control Conference (ACC25)*, 2025
15. McKenzie D, Wu Fung S, Heaton H. Differentiating Through Integer Linear Programs with Quadratic Regularization and Davis-Yin Splitting, *Transactions on Machine Learning Research*, 2024
16. McKenzie D, Heaton H, Li Q, Wu Fung S, Osher S, Yin W. Three-Operator Splitting for Learning to Predict Equilibria in Convex Games. *SIAM Journal on Mathematics of Data Science*, 6 (3), 627-648, 2024
17. Heaton H*, Wu Fung S*. Explainable AI via Learning to Optimize, *Scientific Reports*, 13 (10103), 2023
18. Osher S*, Heaton H*, Wu Fung S*. A Hamilton-Jacobi-based Proximal Operator, *Proceedings of the National Academy of Sciences*, 120 (14), 2023
19. Vidal A, Wu Fung S, Tenorio L, Osher S, Nurbekyan L. Taming Hyperparameter Tuning in Continuous Normalizing Flows Using the JKO Scheme, *Scientific Reports*, 13 (4501), 2023.
20. Heaton H, Wu Fung S, Osher S. Global Solutions to Nonconvex Problems by Evolution of Hamilton-Jacobi PDEs, *Communications on Applied Mathematics and Computation*, 1–21, 2023
21. Chow YT, Wu Fung S, Liu S, Nurbekyan L, Osher S. A Numerical Algorithm for Inverse Problem from Partial Boundary Measurement Arising from Mean Field Game Problem, *Inverse Problems*, 39(1), 014001, 2022
22. Ye J[†], Wan C[†], Wu Fung S. Adaptive Uncertainty-Weighted ADMM for Distributed Optimization, *Journal of Applied and Numerical Optimization*, 4(2), pp. 273-290. 2022
23. Agrawal S, Lee W, Wu Fung S, Nurbekyan L. Random Features for High-Dimensional Nonlocal Mean-Field Games, *Journal of Computational Physics*, 459, pp. 111136. 2022
24. Onken D, Nurbekyan L, Li X, Wu Fung S, Osher S, Ruthotto L. A Neural Network Approach for High-Dimensional Optimal Control, *Transactions on Control Systems Technology*, 31(1), 235-251, 2022
25. Wu Fung S*, Heaton H*, McKenzie D, Li Q, Osher S, Yin W. JFB: Jacobian-free Backpropagation for Implicit Networks, *AAAI Conference on Artificial Intelligence*, 36(6), 6648-6656, 2022
26. Heaton H*, Wu Fung S*, Lin AT*, Osher S, Yin W. Wasserstein-based Projections with Applications to Inverse Problems, *SIAM Journal on Mathematics of Data Science*, 40(2), 581-603, 2022

*denotes co-first author

[†]undergraduate student at time of publication

27. Heaton H*, Wu Fung S*, Gibali A, Yin W. Feasibility-based Fixed Point Networks, *Fixed Point Theory and Algorithms for Sciences and Engineering*, 21, 2021
28. Kan K, Wu Fung S, Ruthotto L. PNKH-B: A Projected Newton-Krylov Method for Large-Scale Bound-Constrained Optimization, *SIAM Journal on Scientific Computing*, 43(5), S704–S726, 2021
29. Lin AT*, Wu Fung S*, Li W, Nurbekyan L, Osher S. Alternating the Population and Agent Control via Two Neural Networks to Solve High-Dimensional Stochastic Mean Field Games, *Proceedings of the National Academy of Sciences*, 118(31). 2021
30. Onken D, Nurbekyan L, Li X, Wu Fung S, Osher S, Ruthotto L. A Neural Network Approach Applied to Multi-Agent Optimal Control, *European Control Conference 2021 (ECC21)*, pp. 1036-1041. 2021
31. Onken D, Wu Fung S, Li X, Ruthotto L. OT-Flow: Fast and Accurate Continuous Normalizing Flows via Optimal Transport, *AAAI Conference on Artificial Intelligence*, 35(10), 9223-9232, 2021
32. Ruthotto L, Osher S, Li W, Nurbekyan L, Wu Fung S. A Machine Learning Framework for Solving High-Dimensional Mean Field Game and Mean Field Control Problems, *Proceedings of the National Academy of Sciences*, 117(17), 2019-22204, 2020 [†]
33. Wu Fung S, Tyrväinen S, Ruthotto L, Haber E. ADMM-Softmax: An ADMM Approach for Multinomial Logistic Regression, *Electronic Transactions on Numerical Analysis*, 52, 214-229, 2020
34. Wu Fung S, Di Z. Multigrid Optimization for Large-Scale Ptychographic Phase Retrieval, *SIAM Journal on Imaging Sciences*, 13(1), 214–233. 2020
35. Wu Fung S, Ruthotto L. An Uncertainty-Weighted Asynchronous ADMM Method for Large-Scale PDE Parameter Estimation, *SIAM Journal on Scientific Computing*, 41(5), S129-S148, 2019
36. Wu Fung S, Ruthotto L. A Multiscale Method for Model Order Reduction in PDE Parameter Estimation, *Journal of Computational and Applied Mathematics*, 350, 19-34, 2019

Refereed Workshop Papers

1. Di N, Chi E, Wu Fung S. A Monte Carlo Approach for Nonsmooth Convex Optimization via Proximal Splitting Algorithms. *NeurIPS Workshop on Optimization for Machine Learning*, 2025
2. Ivanitskiy, MI, Spies AF, Räuker T, Corlouer G, Mathwin C, Quirke L, Rager C, Shah R, Valentine D, Diniz-Behn C, Katsumi I, Wu Fung S. Structured World Representations in Maze-Solving Transformers. *NeurIPS Workshop on Unifying Representations in Neural Models*, 2023.

[†]Author contributions: L.R., S.J.O., W.L., L.N., and S.W.F. designed research; L.R., L.N., and S.W.F. performed research; and L.R., S.J.O., W.L., L.N., and S.W.F. wrote the paper.

Miscellaneous

- Wu Fung S, McKenzie D, Yin W. Learning to Optimize: Where Deep Learning Meets Optimization and Inverse Problems. *SIAM News* 2022.

Honors and Recognition

- 2024 Laney Early Career Alumni Award
 - Awarded annually to a recent graduate of the Laney Graduate School program who has distinguished themselves in service to their fields of endeavor, to Emory, and to society in general.
- Plenary Speaker at the 2024 Georgia Scientific Computing Symposium, Feb 24, 2024
- 2022 MGB-SIAM Early Career (MSEC) Fellow
 - Awarded annually to 8 early career mathematicians for their contributions to the field.
- 2019 Emory Graduate Student Research Award
 - Awarded annually to one graduating PhD student for their research achievements

Contributed and Invited Research Presentations

- *End-to-End Training of High-Dimensional Optimal Control with Implicit Hamiltonians via Jacobian-Free Backpropagation*
 - invited talk at the Optimal Control and Decision Making Under Uncertainty for Digital Twins workshop at the Institute for Mathematical and Statistical Innovation (IMSI). October 23, **2025**
 - invited talk at the International Conference on Continuous Optimization. July 22, **2025**
- *Ensuring Real-Time Safety of Swarm Dynamics via Mean-Field Control Barrier Functions*
 - invited talk at SIAM Conference on Dynamical Systems. May 12, **2025**
 - invited poster at Dynamic Days US. January 4, **2025**
 - invited talk at SIAM Central States Section Annual Meeting 2024. October 6, **2024**
 - invited talk at SIAM Conference on Mathematics of Data Science. October 22, **2024**
- *Kernel Expansions for High-Dimensional Mean-Field Control with Non-local Interactions*
 - invited talk at American Control Conference 2025. July 9, **2025**
- *Explainable AI via Learning to Optimize*
 - Invited Talk at AI Initiative Seminar Series, Oak Ridge National Laboratory. Nov 20, **2025**
 - Invited Talk at AMS Math Club, Colorado School of Mines. Oct 16, **2025**
 - invited talk at Nonlinear Waves Seminar, University of Colorado Boulder. April 8, **2025**
 - invited talk at AMS Colloquium, Colorado School of Mines. February 28, **2025**

- invited talk at SIAM Central States Section Annual Meeting 2024. October 5, **2024**
- invited talk at the CSU Applied Math/Data Science/Inverse Problems Seminar. September 19, **2024**
- invited talk at the 7th Workshop on Autonomous Energy Systems (AES). September 5, **2024**
- invited talk at the Insitute for Pure and Applied Mathematics Computational Microscopy Reunion Conference. June 10, **2024**
- plenary speaker at the 2024 Georgia Scientific Computing Symposium, Emory University. Feb 24, **2024**
- invited talk at the Level Set Collective Seminar, UCLA. Dec 4, **2023**
- invited talk at the Mathematical Biology Research Group Seminar, Colorado School of Mines. October 19, **2023**
- *Using Hamilton Jacobi PDEs in Optimization*
 - invited talk at Data-Driven Methods for Science and Engineering Seminar, University of Washington. April 7, **2023**
 - invited talk at the Mathematical Machine Learning Seminar, Max Planck Institute. March 2, **2023**
 - invited talk at the Center for Mathematics and Artificial Intelligence, George Mason University. February 24, **2023**.
 - invited talk at the Center for Research in Signals and Networks Seminar, Colorado School of Mines. February 11, **2023**.
 - invited talk at the Applied and Computational Mathematics Division Seminar Series. National Institute of Standards and Technology. Boulder, CO, January 24, **2023**
- *Global Solutions to Nonconvex Problems by Evolution of Hamilton-Jacobi PDEs*
 - invited talk at the Spatial Statistics and Kernel Club. Colorado School of Mines. Golden, CO, October 12, **2022**
 - invited talk at the Optimal Transport and Mean Field Game Seminar at University of South Carolina. April 7, **2022**.
 - invited talk at Level Set Collective Seminar, Department of Mathematics, UCLA, Los Angeles, Ca. September, 20, **2022**
 - invited talk at the Early Career Math Colloquium at The University of Arizona. September 21, **2022**.
- *A Deep Learning Approach for Real-Time High-Dimensional Optimal Control*
 - invited talk at the 4th AFOSR Monterey Training Workshop on Computational Issues in Nonlinear Control. May 24, **2023**
 - invited talk at Colorado School of Mines, Math Club/SIAM Student Chapter. March 16, **2022**.
 - invited talk at the Center for Research in Signals and Networks Seminar, Colorado School of Mines. February 18, **2022**.

- invited talk at the Hamilton-Jacobi PDEs Reunion Conference I, at the Institute for Pure and Applied Mathematics, Los Angeles, California, January 13, **2022**.
- *Efficient Training and Design of Implicit Networks with Applications in Contextual Games*
 - invited talk at the Center for Research in Signals and Networks Seminar, Colorado School of Mines. December 10, **2021**.
 - invited talk at SIAM Conference on Optimization. Seattle, Wa. May 31, **2023**
- *Efficient Training of Infinite-depth Neural Networks via Jacobian-free Backpropagation*
 - invited talk at Sacred Heart University. Fairfield, CT. October 3, **2022**.
 - invited talk at SIAM Conference on Mathematics of Data Science. San Diego, CA. September 30, **2022**.
 - invited talk at the CS@Mines Seminar at Colorado School of Mines. Golden, CO. May 3, **2022**.
 - invited talk at SIAM Conference on Uncertainty Quantification. Atlanta, Ga. April 14, **2022**.
 - invited talk at the Math Colloquium Series, at University of Colorado, Colorado Springs, March 31, **2022**.
 - invited talk at the Los Alamos National Lab ML Seminar, February 17, **2022**.
 - invited talk at the The Carl Heiland Lecture Series, at the Department of Geophysics, Colorado School of Mines, February 9, **2022**.
 - invited talk at the Applied Math/Inverse Problems Seminar, at Colorado State University, February 3, **2022**.
 - invited talk at the The Scientific AI Research Group, at the University of Texas at Austin, January 28, **2022**.
 - invited talk at the Center for Wave Phenomena Seminar, Colorado School of Mines. December 6, **2021**
 - invited talk at the AMS Fall Western Sectional Meeting, at University of New Mexico. October 23, **2021**.
 - invited talk at the PDE and Applied Math Seminar at the University of California, Riverside. October 20, **2021**.
 - invited talk at the Statistics, Optimization and Machine Learning Seminar at University of Colorado, Boulder. October 12, **2021**.
 - contributed talk at the Applied Math and Statistics Colloquium at Colorado School of Mines. September 10, **2021**.
- *Wasserstein-based Projections for Inverse Problems*
 - invited talk at the Applied and Computational Mathematics Seminar at Dartmouth College. January 26, **2021**.
 - invited talk at the PDE and Applied Math Seminar at the University of California, Riverside. January 20, **2021**.

- invited talk at the Deep Learning Seminar at University of South Carolina. December 1, **2020**.
- invited talk at the Optimal Transport and Mean Field Game Seminar at University of South Carolina. October 14, **2020**.
- invited talk at the Mathematics and Deep Learning Collective at Iowa State University. October 2, **2020**.
- *A GAN-based Approach for High-Dimensional Stochastic Mean Field Games*, held at
 - invited talk at the Spatial Statistics and Kernel Club. Colorado School of Mines. Golden, CO, March 11, **2022**
 - invited talk at the SIAM Virtual Conference on Mathematics of Data Science. June 25, **2020**
 - invited talk at the Laboratory for Applied Mathematics, Numerical Software, and Statistics (LANS) Seminar at Argonne National Laboratory. June 17, **2020**.
 - invited talk at the Numerical Analysis and Scientific Computing Seminar at Emory University. Atlanta, Ga. March 27, **2020**.
- *A Machine Learning Framework for High-Dimensional Mean Field Games*, held at
 - invited talk at the Optimal Transport and Applications to Machine Learning and Statistics workshop at MSRI, Berkeley, Ca, May 5, **2020**
 - invited talk (joint with Stanley Osher) at the High Dimensional Hamilton-Jacobi Methods in Control and Differential Games workshop at IPAM, Los Angeles, Ca, April 1, **2020**
 - contributed poster in the Intersections between Control, Learning and Optimization workshop at IPAM, Los Angeles, Ca, February 24, **2020**
 - invited talk at the Level Set Collective Seminar, Department of Mathematics, UCLA, Los Angeles, Ca. December 3, **2019**
- *Adaptive Multiscale and Asynchronous Optimization Methods for Large-Scale PDE Parameter Estimation*, held at
 - invited talk at the Level Set Collective Seminar, Department of Mathematics, UCLA, Los Angeles, Ca. July 30, **2019**
 - invited talk at AMS Spring Southeastern Sectional Meeting, Auburn, AL, March 17, **2019**
 - invited talk at SIAM Conference on Computational Science and Engineering, Spokane, Wa. February 27, **2019**.
- *Large-Scale Classification using Multinomial Regression and ADMM*
 - contributed poster at Georgia Scientific Computing Symposium. Atlanta, Ga. February 16, **2019**
- *Multilevel Algorithms for Ptychographic Phase Retrieval*, held at various occasions:
 - contributed talk at the Summer Argonne Student Symposium at Argonne National Laboratory. Lemont, Il. July 26, **2018**

- invited talk at the Advanced Photon Source at Argonne National Laboratory. Lemont, IL. July 16, **2018**
- *An Uncertainty-Weighted ADMM Method for Large-Scale PDE Parameter Estimation*, held at various occasions:
 - invited talk at SIAM Conference on Uncertainty Quantification. Garden Grove, Ca. April 19, **2018**
 - contributed talk at Fifteen Copper Mountain Conference on Iterative Methods. Copper Mountain, Co. March 26, 2018
 - invited talk at Spelman College. Atlanta, Ga, February 26, **2018**
 - contributed poster at Georgia Scientific Computing Symposium. Atlanta, Ga. February 24, **2018**
 - contributed talk at the Scientific Computing Seminar at Emory University. Atlanta, Ga, USA, October 13, **2017**
- *jInv - A Flexible Julia Package for Parallel PDE Parameter Estimation*, held at various occasions:
 - contributed e-poster at SIAM Conference on Computer Science and Engineering, Atlanta, GA, March 1, **2017**
 - contributed poster, Georgia Scientific Computing Symposium. Atlanta, Ga. February 20, 2016
- *PDE-Constrained Optimization with Multiscale Methods*, held at various occasions:
 - invited talk at SIAM Annual Meeting Conference. Pittsburgh, Pa, USA. July 10 - 14, **2017**
 - invited talk at SIAM Conference on Computational Science and Engineering. Atlanta, Ga, USA, March 3, **2017**
 - contributed talk at the Scientific Computing Seminar at Emory University. Atlanta, Ga, USA, February 17, **2017**

Teaching

- MATH 307: Intro to Scientific Computing, Colorado School of Mines
 - Fall 2025, Spring 2024, Fall 2023, Fall 2022, Fall 2021
- MATH 521 / EENG521: Numerical Optimization (Graduate Level), Colorado School of Mines
 - Spring 2026, Spring 2024, Spring 2023
- MATH 398: Introduction to Mathematical Optimization, Colorado School of Mines
 - Spring 2025
- MATH 598B: Mathematical Foundations of Interpretability and Alignment for LLMs (Graduate Level), Colorado School of Mines

- Spring 2026, Spring 2025, Spring 2024
- MATH 270C: Computational Linear Algebra (Graduate Level), UCLA
 - Winter 2021
- MATH 151A: Applied Numerical Methods I UCLA
 - Spring 2021, Spring 2020, Fall 2019
- MATH 151B: Applied Numerical Methods II, UCLA
 - Fall 2020, Winter 2020
- MATH 111: Introductory Calculus, Emory University
 - Fall 2016, Spring 2016, Fall 2015

Graduate Advisees

- Abigail Broscius (PhD). Project: Deep Learning-based AC Optimal Power Shutoff for Power-Grid-Related Wildfire Mitigation. Co-advised with Qihua Huang, Colorado School of Mines, since August 2025. **Awarded an NSF Graduate Research Fellowship.**
- Eric Gelphman (PhD). Project: Implicit Deep Learning for Optimal Control. Colorado School of Mines, since August 2024.
- Andrew Holmberg (PhD). Project: Scalable Algorithms for Decision-Focused Learning. Colorado School of Mines, since August 2024.
- Kaitlin Raitz (MS). Project: Effects of Stochasticity on Jacobian-Free Backpropagation. Co-advised with Daniel McKenzie, Colorado School of Mines, Aug. 2024 - .
- Brandon Knutson (PhD). Project: Algorithms for Training Implicit Neural Networks and their Logical Extrapolation. Co-advised with Daniel McKenzie, Colorado School of Mines, since January 2024.
- Michael Ivanitsky (PhD). Project: Mechanistic Interpretability of Maze-Solving Transformers. Co-advised with Cecilia Diniz-Behn, Colorado School of Mines, Jan. 2022 - May 2026. **Awarded the Colorado School of Mines Rath Award.** Now a Research Fellow at the ML Alignment & Theory Scholars (MATS) program.
- Soraya Terrab (PhD). Project: Data-Driven Multiwavelet Methods for Discontinuity Detection. Co-advised with Jennifer Ryan, Colorado School of Mines, Jan. 2022 - Feb 2025. Now an Imaging Science Engineer at Ricoh USA, Inc.
- Alexander Vidal (PhD). Project: Deep Learning Methods for Large-Scale Physics, Sept. 2022 – May 2024. Now a Sr. Machine Learning Researcher at Launch Potato.

Undergraduate Advisees

- Jordan Pettyjohn. Project: *Logical Extrapolation via Implicit Deep Learning*. Colorado School of Mines, August 2022 - May 2025. Now a PhD student in Computer Science at University of Chicago
- Amandin Chyba. Project: *Logical Extrapolation via Implicit Deep Learning*. Colorado School of Mines, August 2022 - May 2023. Now a PhD student in Mathematics at the Courant Institute at NYU.
- Ibrohim Nosirov. Project: *Deep Learning Methods for Signal Processing*. Colorado School of Mines, September 2021 - December 2021. Co-advised with Mike Wakin. Now a PhD student in Computational and Applied Mathematics at Cornell University.
- Caleb Wan. Project: *Adaptive Uncertainty-Weighted ADMM Methods for Machine Learning*. UCLA, July 2020 - December 2021.
- Jiangping Ye. Project: *Adaptive Uncertainty-Weighted ADMM Methods for Machine Learning*. UCLA, July 2020 - December 2021. Now a PhD student in applied mathematics at the University of Maryland.
- Richard Yim. Project: *Learned Inverse Scale Space Flows*. UCLA, January 2020 - June 2020. Now a Bioinformatics Research Analyst at UCSF.
- Sudhanshu Agrawal. Project: *Machine Learning for High-Dimensional Non-Local Mean Field Games*. UCLA, January 2020 - February 2022. Co-advised with Levon Nurbekyan. Now a Machine Learning Engineer at Qualcomm.
- Emory 2022 REU/RET Program on Model Meets Data. Project: Implicit Deep Learning for Inverse Problems.
 - Linghai Liu, Brown University. Now a PhD student in Statistics at Yale University
 - Allen Tong, UCLA. Now a PhD student in Applied Mathematics at Georgia Tech.
 - Lisa Zhou, UC Berkeley.
- Research in Industrial Projects for Students (RIPS). Institute for Pure and Applied Mathematics, UCLA. June 2020 - Aug 2020. Project: *Large-Scale Inventory Optimization*
 - Miranda Kaiser, Rensselaer Polytechnic Institute.
 - Julia Balukonis, Providence College.
 - Rachel Fan, Vanderbilt University
 - Rong (Hugh) Jiang, UC Berkeley

Graduate Thesis Committees

(My own students are listed above)

- Sweta Rai, PhD, Colorado School of Mines. July 2025
- Meng Jia, PhD, Colorado School of Mines. May 2024
- Kylee Shiekh, MSc, Colorado School of Mines. January 2024

- Will Daniels, PhD, Colorado School of Mines. May 2024
- Matthew Picklo, PhD, Colorado School of Mines. November 2023
- Austin Hoyle, MSc, Colorado School of Mines. December 2023

Other Skills

- Programming Languages: Python, Julia, Matlab
- Languages: Spanish (native), English (fluent), French (fluent), Cantonese (fluent)

Conference, Seminar, and Minisymposium Organization

○ Conference Organization

- Organizing Committee Member of the 2026 SIAM Conference on Mathematics of Data Science

○ Seminar Organization

- Co-organizer of Mines Optimization and Deep Learning Seminar, Colorado School of Mines
- Co-organizer of Applied Mathematics and Statistics Colloquium, Colorado School of Mines
- Organizer of Kernel Club Seminar, Colorado School of Mines

○ Minisymposium Organization

- Co-organizer of mini-symposium on *Theory and Applications of Deep Equilibrium Networks* at 2024 INFORMS Optimization Society Conference, Houston, Texas. March 2024
- Co-organizer of mini-symposium on *Addressing Intractability in Optimal control* at SIAM Conference on Computational Science and Engineering, 2025 in Fort Worth, Texas. March 2025
- Co-organizer of mini-symposium on *Advances in Optimization and Feasibility Methods for and with Machine Learning* at SIAM Conference on Optimization, Seattle, Washington. May 2023
- Co-organizer of mini-symposium on *Advances in Learning to Optimize and Optimizing to Learn* at SIAM Conference on Mathematics of Data Science, San Diego, California. September 2022
- Co-organizer of mini-symposium on *Deep Learning Methods for Optimization* at SIAM Conference on Uncertainty Quantification, Atlanta, Georgia, USA. April 2022
- Co-organizer of mini-symposium on *Advances in Regularization Techniques for Ill-Posed Problems* at the SIAM Conference on Imaging Sciences, Toronto, Canada. July, 2020
- Co-organizer of mini-symposium on *Advances in Optimal Control for and with Machine Learning* at the SIAM Conference on Mathematics of Data Science, Cincinnati, Ohio. May, 2020

- Co-organizer of mini-symposium on *Mathematical Advances in Deep Learning* at the SIAM Conference on Computational Science and Engineering, Spokane, Washington. February, 2019

Professional Activities and Affiliations

- Executive Guest Editor for Special Issue on Optimal Transport in the Journal of Computational Physics. 2026
- Member of the American Mathematical Society (AMS)
- Member of the Society for Industrial and Applied Mathematics (SIAM)
- Reviewer for the following journals and conferences:
 - 2025 American Control Conference (ACC25)
 - Physica D: Nonlinear Phenomena
 - SIAM Undergraduate Research Online
 - SIAM Journal on Numerical Analysis
 - SIAM Journal on Scientific Computing
 - SIAM Journal on Imaging Sciences
 - Frontiers in Applied Mathematics and Statistics
 - Mathematical and Scientific Machine Learning Conference (MSML)
 - Journal of Applied and Numerical Optimization (JANO)
 - Inverse Problems
- Co-founder of the Mines Optimization and Deep Learning Research Group.
- Board Member for the Emory SIAM Student Chapter. Aug 2014 - May 2019